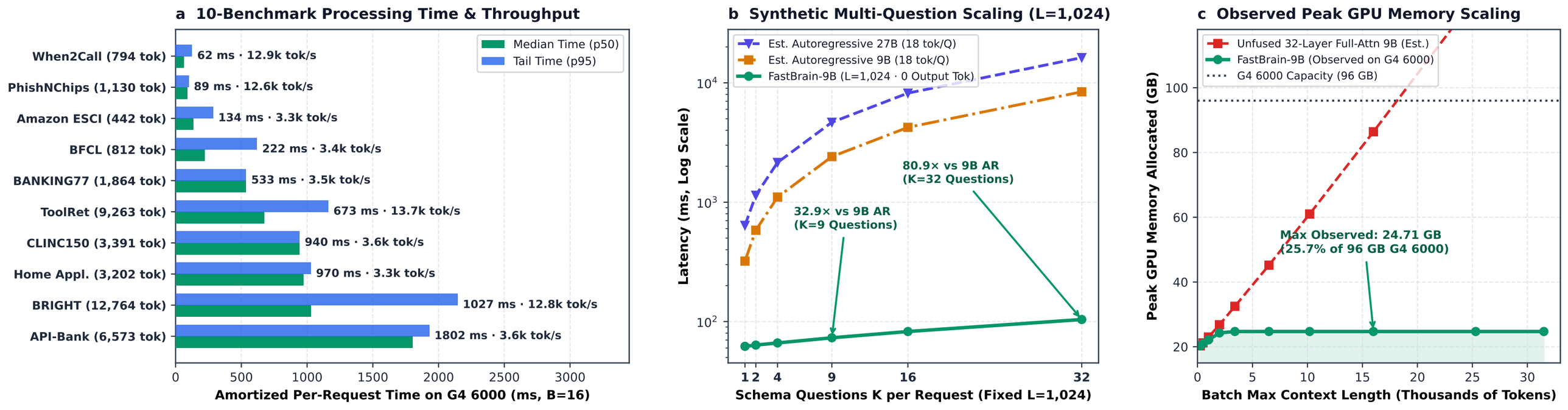
a 10-Benchmark Processing Time & Throughput
Median Time (p50)
Tail Time (p95)
When2Call (794 tok)
62 ms · 12.9k tok/s
PhishNChips (1,130 tok)
89 ms · 12.6k tok/s
Amazon ESCI (442 tok)
134 ms · 3.3k tok/s
BFCL (812 tok)
222 ms · 3.4k tok/s
BANKING77 (1,864 tok)
533 ms · 3.5k tok/s
ToolRet (9,263 tok)
673 ms · 13.7k tok/s
CLINC150 (3,391 tok)
940 ms · 3.6k tok/s
Home Appl. (3,202 tok)
970 ms · 3.3k tok/s
BRIGHT (12,764 tok)
1027 ms · 12.8k tok/s
API-Bank (6,573 tok)
1802 ms · 3.6k tok/s
0
500
1000
1500
2000
2500
3000
Amortized Per-Request Time on G4 6000 (ms, B=16)
b Synthetic Multi-Question Scaling (L=1,024)
Est. Autoregressive 27B (18 tok/Q)
Est. Autoregressive 9B (18 tok/Q)
FastBrain-9B (L=1,024 · 0 Output Tok)
80.9× vs 9B AR
(K=32 Questions)
32.9× vs 9B AR
(K=9 Questions)
Latency (ms, Log Scale)
$10^4$
$10^3$
$10^2$
1 2 4 9 16 32
Schema Questions K per Request (Fixed L=1,024)
c Observed Peak GPU Memory Scaling
Unfused 32-Layer Full-Attn 9B (Est.)
FastBrain-9B (Observed on G4 6000)
G4 6000 Capacity (96 GB)
Max Observed: 24.71 GB
(25.7% of 96 GB G4 6000)
Peak GPU Memory Allocated (GB)
100
80
60
40
20
0
5
10
15
20
25
30
Batch Max Context Length (Thousands of Tokens)